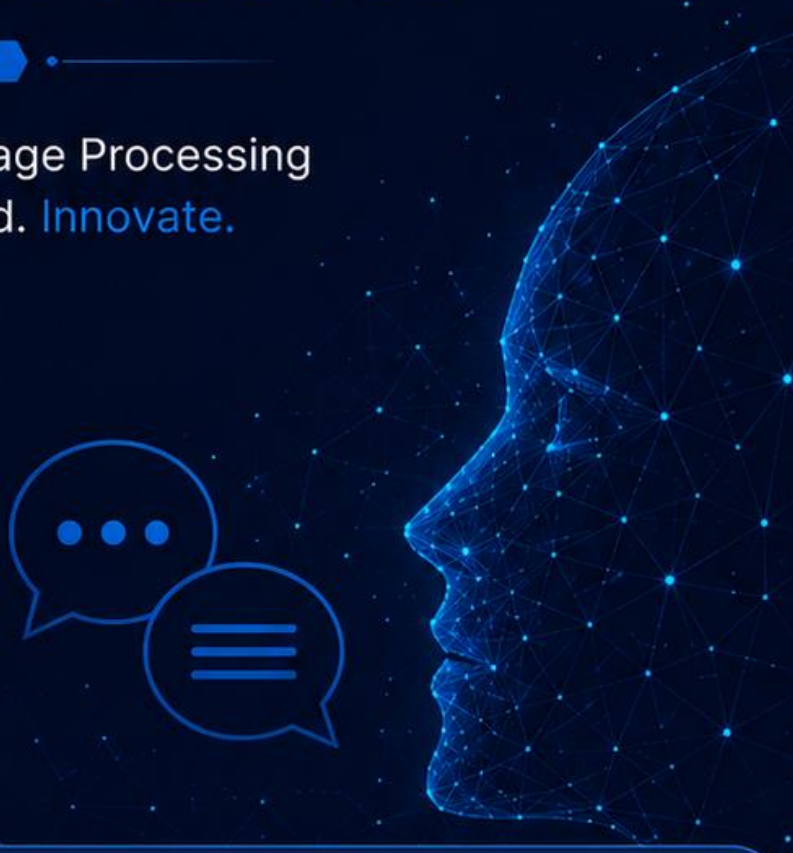
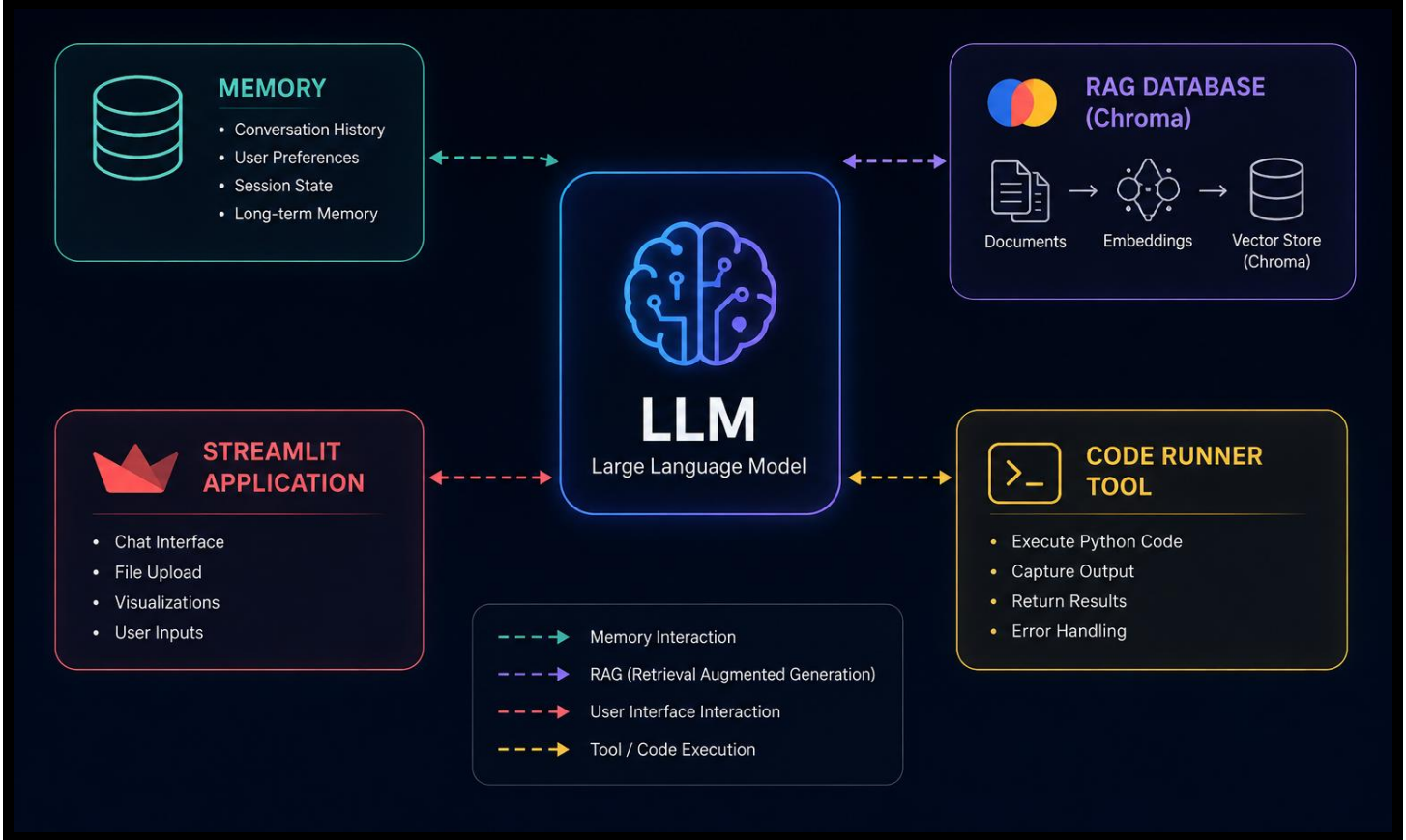
Cellula Technologies

NLP INTERNSHIP

Natural Language Processing
Learn. Build. Innovate.



Project 2


MEMORY
Conversation History
User Preferences
Session State
Long-term Memory
RAG DATABASE (Chroma)
Documents
Embeddings
Vector Store (Chroma)
LLM
Large Language Model
STREAMLIT APPLICATION
Chat Interface
File Upload
Visualizations
User Inputs
CODE RUNNER TOOL
Execute Python Code
Capture Output
Return Results
Error Handling
Memory Interaction
RAG (Retrieval Augmented Generation)
User Interface Interaction
Tool / Code Execution

Voice-Based Data Analysis System

Complete the existing project by transforming it into a fully functional voice-based data analysis system using FastAPI instead of Streamlit.

The main objective is to allow users to analyze their datasets using natural voice commands. The user should be able to upload or provide a dataset and then ask analytical questions by speaking naturally, without manually writing SQL queries.

The system should use Faster-Whisper to convert the user's speech into text. The transcribed text is then passed to an LLM, which interprets the user's request and generates the appropriate SQL query based on the structure and contents of the dataset.

The generated SQL query must be validated and executed using Python's built-in SQLite3 database engine. The system should return the query result in a clear and useful format. Whenever appropriate, the system should also provide meaningful explanations of the results.

The complete application must be implemented using FastAPI as the backend API framework rather than Streamlit. Students should design appropriate API endpoints for file/data input,

audio upload, speech-to-text conversion, LLM-based SQL generation, SQL execution, and returning analysis results.

The final system should demonstrate a complete pipeline:

Voice → Faster-Whisper → Text → LLM → SQL Query → SQLite3 → Data Analysis Result

Important Note : Can Support Arabic voice, not English voice only , which mean it can convert Arabic text to SQL Query too

1 User Voice Input





"Show me total sales by product category"

The user speaks a question or request in natural language.

2 Speech-to-Text (Faster-Whisper)



faster whisper

"Show me total sales by product category"

Faster-Whisper model converts speech to text.

3 LLM (SQL Generation)

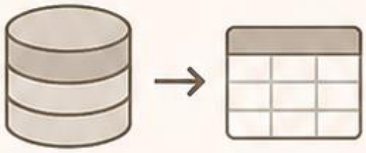


```
SELECT category,
SUM(sales) as total_sales
FROM sales
GROUP BY category;
```

The LLM understands the request and generates the appropriate SQL query.

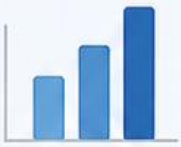
4 Execute SQL (SQLite3)





The SQL query is executed on the SQLite3 database.

5 Data Analysis Result



category	total_sales
Electronics	125000
Clothing	87000
Home & Kitchen	54000
Books	23000

The result is returned to the user in a clear and useful format.